

Technology Validation and Start Up Fund

Round 31 Proposal Evaluations

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Submitted to:
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1) Executive Summary

Redwood is a Columbus, Ohio based LLC founded by former Battelle executives over 9 years ago. Redwood has assembled an extraordinary team for this Program. Each member of the five-person Redwood team is an accomplished technology commercialization professional with decades of experience in performing business and technical evaluations. This team, combined with identified external subject matter experts, has extensive experience in all six of the Ohio Third Frontier technology focus areas. More detail on the Redwood team is provided in Appendix 1 of this report and on our website (www.Redwdinnnov.com). Details of the TVSF program and the review process are provided in Appendix 2.

Three (3) TVSF Round 31 Phase 2 applications were received and initially reviewed. One application was rejected without further review because it was submitted by an ineligible applicant and is not included in the analytic results. There were no Phase 1 applications this round. The remaining 2 Phase 2 applications totaled \$250,000. Funding is recommended for 1 Phase 2 application for a total of \$100,000. Funding is not recommended for 1 Phase 2 application for a total of \$150,000. This translates to a 50% recommended application funding rate for this TVSF round, compared to the average of 48% over all 31 TVSF rounds.

2) Evaluation Results

Summaries of the evaluations of the proposals and funding recommendations are shown in Table 1. Questions were submitted to applicants to answer prior to conducting video interviews. The total recommended funding for Phase 2 projects is \$100,000. Note that the Table 1 column widths are proportional to the weighting of the evaluation criteria. For example, in Table 1, Management Team which is weighted at 20 is four times as wide as ESP Interaction which is weighted at 5. Note that a yellow evaluation indicates that the proposal meets that particular criterion.

More detailed evaluations and recommendations for each Phase 2 proposal may be found in Section 3 of this report.

TECHNOLOGY VALIDATION AND STARTUP FUND

Table 1 – Phase 2 Proposal Evaluation and Funding Recommendation

Phase 2 Proposal Evaluation and Funding Recommendation TVSF Round 31

Proposal Number	Lead Applicant	Requested Funding (\$1000)	Recommended Funding (\$1000)	Team	Opportunity & Market Size	IP Protection	Proof of Concept	Potential Invest or & Business Partner Engagement	Business Model	Project Plan & Budget Narrative	Growth Plan in Ohio	ESP Interaction
FY23-3764	Smart Response Technologies	100	100									
	Sub-Total	100	100									
FY23-3767	Neuright, Inc.	150	-									
	Sub-Total	150	-	Column width is proportional to score weighting in each category								
	Total	250	100									
Evaluation Scale				Weak		Meets		Exceeds				

TECHNOLOGY VALIDATION AND STARTUP FUND

Table 2 lists the funding approval rate by TVSF round. This round's approval rate is 50% of the total reviewed proposals. The historical range of individual rounds has spanned 27 – 100%, with an average of 48%.

Table 2. TVSF Approval Rate by Round

TVSF Round 31					
Approval Rate by Round					
Phase 1/Phase 2					
Round	\$ Recommended	Approval Rate	Round	\$ Recommended	Approval Rate
1 (APR 2012)	\$950,000	35%	20 (NOV 2019)	\$1,350,000	43%
2 (AUG 2012)	\$900,000	52%	21 (FEB 2020)	\$3,944,000	56%
3 (DEC 2012)	\$610,000	44%	22 (JUN 2020)	\$1,398,630	53%
4 (JUN 2013)	\$864,000	30%	23 (DEC 2020)	\$900,000	50%
5 (FEB 2014)	\$1,462,000	46%	24 (MAR 2021)	\$2,092,900	55%
6 (JUN 2014)	\$998,000	39%	25 (JUN 2021)	\$800,000	75%
7 (OCT 2014)	\$1,100,000	57%	26 (OCT 2021)	\$1,700,000	55%
8 (FEB 2015)	\$710,000	37%	27 (FEB 2022)	\$850,000	43%
9 (JUN 2015)	\$550,000	31%	28 (APR 2022)	\$2,499,976	64%
10 (DEC 2015)	\$925,000	38%	29 (JULY 2022)	\$850,000	100%
11 (APR 2016)	\$1,239,000	46%	30 (OCT 2022)	\$3,700,000	71%
12 (OCT 2016)	\$3,537,269	46%	31 (JAN 2022)	\$100,000	50%
13 (MAR 2017)	\$1,567,500	38%	Overall	\$45,945,707	
14 (SEP 2017)	\$498,832	27%	Average	1,531,524	48%
15 (DEC 2017)	\$2,250,000	38%			
16 (MAR 2018)	\$2,098,600	52%			
17 (SEP 2018)	\$2,100,000	42%			
18 (DEC 2018)	\$1,150,000	35%			
19 (APR 2019)	\$2,250,000	43%			

3) Proposal Summary

Proposal Summaries - Phase 2 Recommended for Funding

Proposal 23-3764	Smart Response Technologies	Amount Requested: \$100,000
<i>Licensing Institution</i>	Air Force Research Laboratory	Amount Recommended: \$100,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: 30	<i>Delphini</i>

Company Snapshot: Automatic Speech Recognition Software-as-Service Package for English & Spanish transcriptions of phone conversations in Public Safety Applications including correctional institutions, and 911 and 988 calls.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Management Team	Management team has expertise in software solutions, military sales, public safety, program management, general business. Management team 100% committed to project and company operations. President is based in Ohio. Majority of future hiring is expected to be in Ohio.
Y	Opportunity/Market Size	Initial market focus is transcriptions of phone conversations for public safety markets including correctional facilities, 911 & 988 emergency calls. TAM is \$117 MM for these markets.
Y	Intellectual Property Protection	AFRL USP 9,230,549 - Multi-Modal Communication, filed May 2011. Patent license has been granted in relevant field-of-use. Confidential know-how developed, providing competitive advantage.
Y	Compelling Proof of Concept	Based on the Delphini™ ASR (TRL 7) repurposed emergency radio acoustic/language model for correctional facility. Piloting at Lehigh County Jail, July 2022 allowed SRT to build acoustic model.
Y	Potential Investor/ Business Partner Engagement	Engaged with potential customers. Tetras is sale channel partner for correctional facilities. Ongoing outreach for seed round funding.
Y	Business Model	Subscription ~ \$40,000/year. Additional revenue for add-on features. Revenue ~\$11 MM (Yr 5). Develop/support strategic sales channels.
Y	Project Plan/ Budget Narrative	Project plan defined. Strong contractors identified. Budget confirmed.
Y	Growth Plan in Ohio	Plan is to operate and hire personnel in Dayton Ohio Area.
G	ESP Interaction	EC of Dayton active participation in project & company.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Management team has strong commitment and relevant technical and commercial expertise to successfully complete project and grow company in Ohio. Focus, timing, and rational of market segment development is explained in a logical and credible manner. Competitive position is provided by licensed USAF patent plus significant existing and anticipated in-house know-how and expertise. Business model relies on maintaining "best-in-class" product performance and successful direct and outside strategic sales channels. Significant existing technology platform from prior work on emergency command and control center (CCC) applications. Ongoing outreach to seed funding sources, strategic sales channel partners, and direct customers.

Proposal Summary - Phase 2 Not Recommended for Funding

Proposal 23-3767	Neuright, Inc.	Amount Requested: \$150,000
<i>Licensing Institution</i>	University of Maine	Amount Recommended: \$0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	<i>First in human validation of a novel medical diagnostic device for peripheral neuropathy</i>

Company Snapshot: Neuright is developing a device to diagnose peripheral neuropathy (PN).

Subsequent versions of the device will be developed to treat PN.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	Highly qualified technical team. Consultants, support from Rev1, outside service providers and experts would support the team sufficiently. Core team will add medical device development or corporate experience.
Y	Opportunity/Market Size	Peripheral Neuropathy Market is projected to reach \$292.40 million USD by 2025, a 3.9% CAGR. Other estimates place current market above \$1 billion.
Y	Intellectual Property Protection	"Methods and Devices For Treatment of Neuropathy" (PCT/US21/31337). In negotiations for exclusive license. IP protection may not cover all versions of the product. New IP from OSU in development.
G	Proof of Concept	Device needed for clinical testing is designed but must be manufactured in quantity for the clinical studies to be performed with TVSF funds. Outputs are clearly defined and described.
G	Potential Investor/ Business Partner Engagement	Discussions with investors and grant funders indicate need for proof-of-concept clinical data from TVSF-funded trial and other milestones.
R	Business Model	Proforma budget provided but description of business model is scant. Mechanisms or channels to achieve sales are unclear.
G	Project Plan/ Budget Narrative	The project plan is clearly described and achievable in 12 months.
R	Growth Plan in Ohio	Company management in Columbus. Other sources show Maine Co.
Y	ESP Interaction	Support and consultation from Rev1.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: The founder / scientific lead understands the clinical unmet need and how technology should be developed. Budget and aims well organized. Core team lacks medical product or corporate experience. Business support from Rev1, consultants and outside service providers will be needed and should be continued. The business model is not adequately explained. Written answers and interview did not sufficiently clarify how they would approach selling their medical device. Unclear if IP will protect the simplest diagnostic version of the device. New biomarker IP is in development at OSU. Commitment to Ohio location not adequately demonstrated. Online presence shows Maine location and U Maine support letter states Neuright is Maine-based. This application is not recommended for funding.

4) Round 31 Analysis

Figure 1 shows the proposal activity and funding recommendations by technology source for Phase 2 proposals. There was one submission each from Air Force Research Laboratory and the University of Miami. One application is recommended for funding.

Figure 1. Round 31 Funding by Technology Source

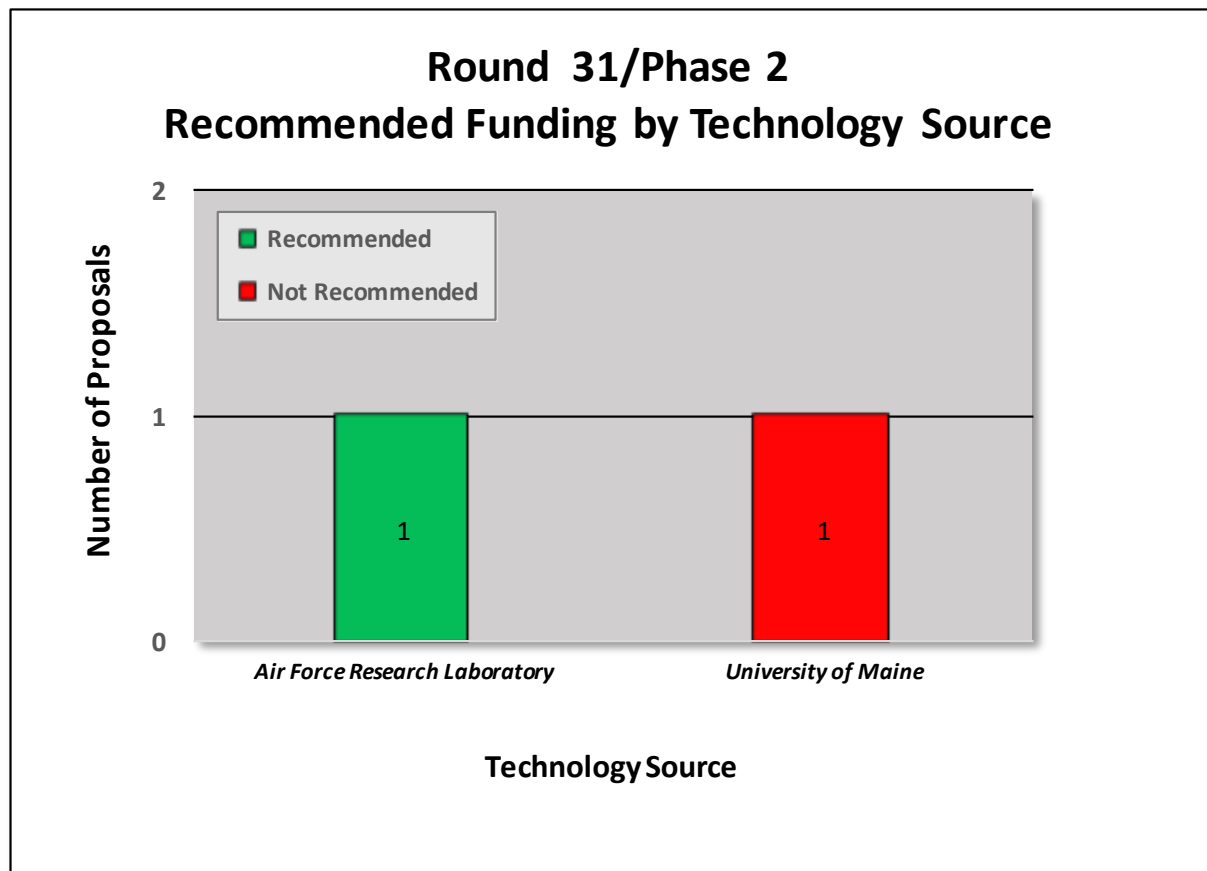
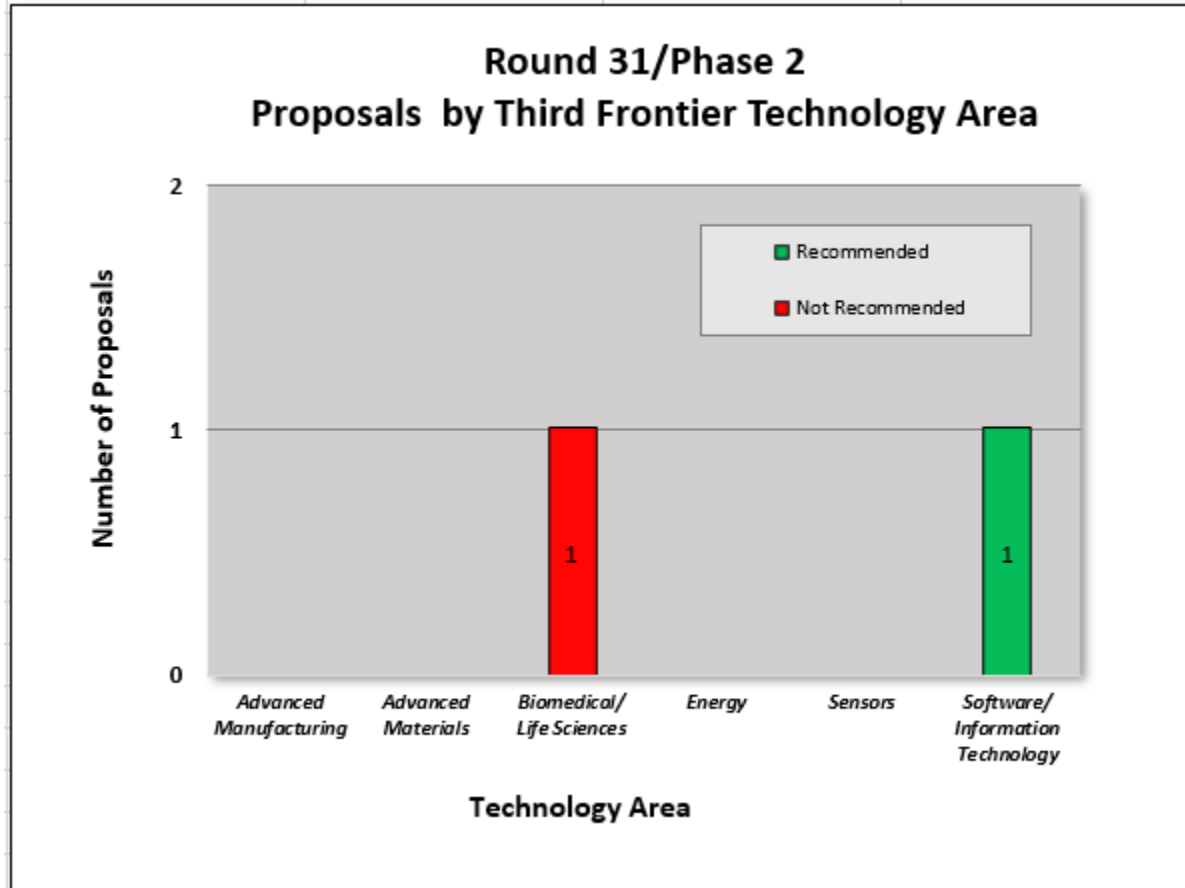


Figure 2 depicts Phase 2 proposal activity and funding recommendations by Third Frontier focus area. In Round 31, one of two proposals (50%) is in Biomedical/Life Sciences, one of two is in Software/Information Technology (50%), the latter of which is recommended for funding.

Figure 2. Round 31 Phase 2 Proposal Activity by Third Frontier Technology Area



TECHNOLOGY VALIDATION AND STARTUP FUND

Figure 3 shows the aggregate ratings by evaluation criteria for all Phase 2 proposals. Team, Proof of Concept, Investor/Partner, Project Plan Budget and ESP interaction were the strongest categories in Round 31. Business Model and Growth Plan in Ohio were rated as the weakest.

Figure 3. Round 31 Phase 2 Proposal Rating Summary

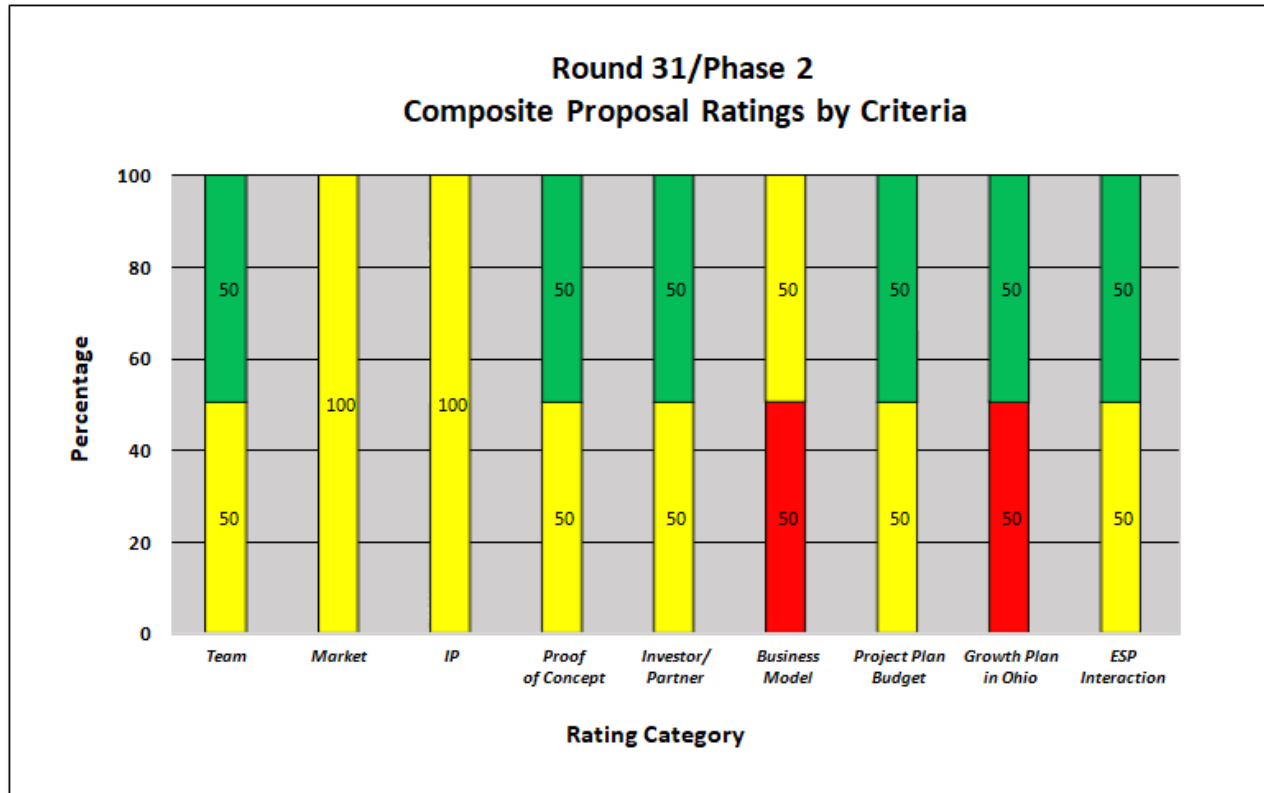


Figure 4 shows the percent meets or exceeds of the business model by Round. In the previous eleven Rounds, business model was the lowest rating in Rounds 20-23 (53% average $\geq$ meets), Round 26 (28%) and Round 31 (50%). The RFP was revised to elicit stronger business models prior to Round 24 and it appears that the proposals have provided stronger business models in subsequent Rounds. The last 8 rounds average 67% average $\geq$ meets, even with the Round 26 (28%). The average over all 12 rounds is 62%.

Figure 4: Rounds 20 to 31 Phase 2 Analysis of Business Model

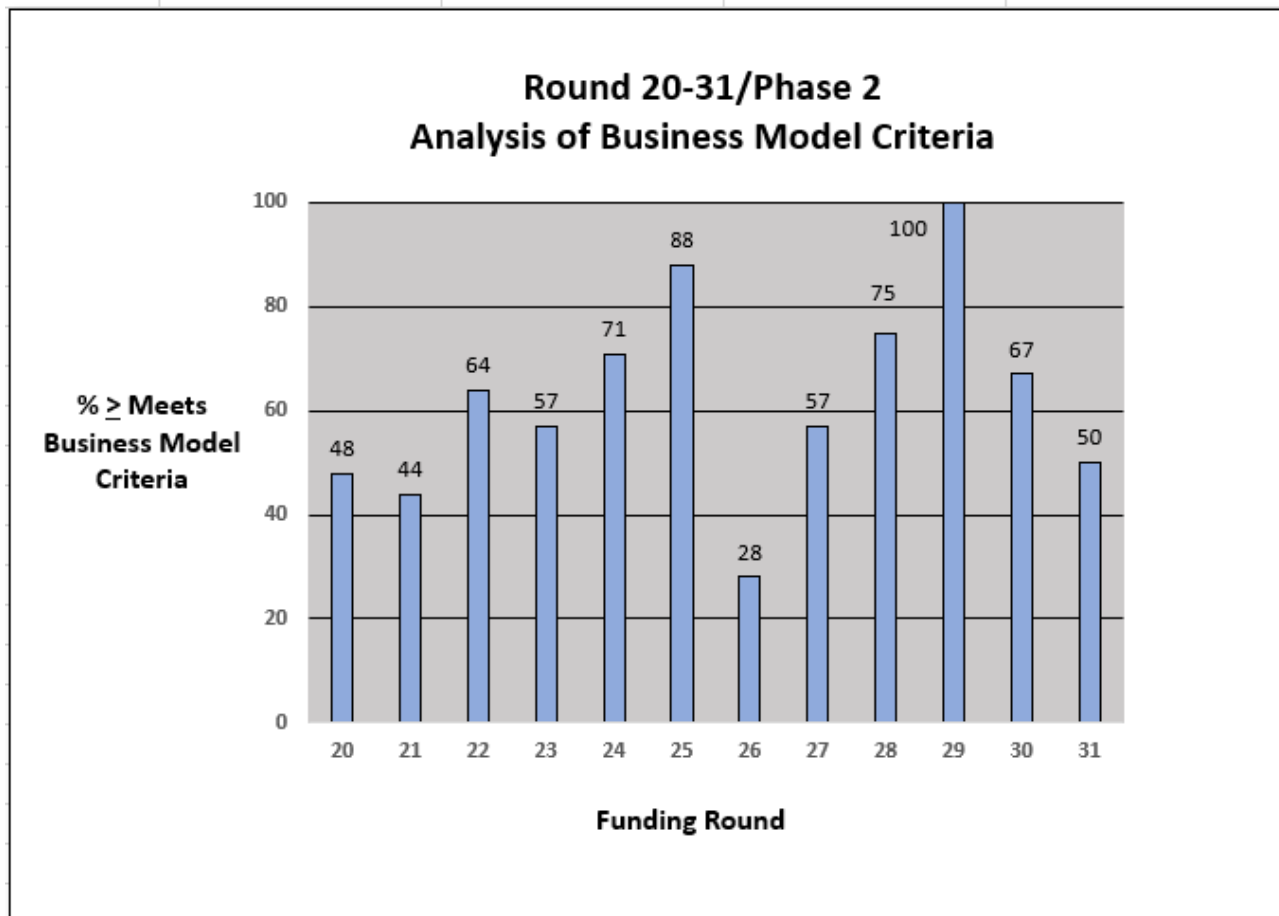
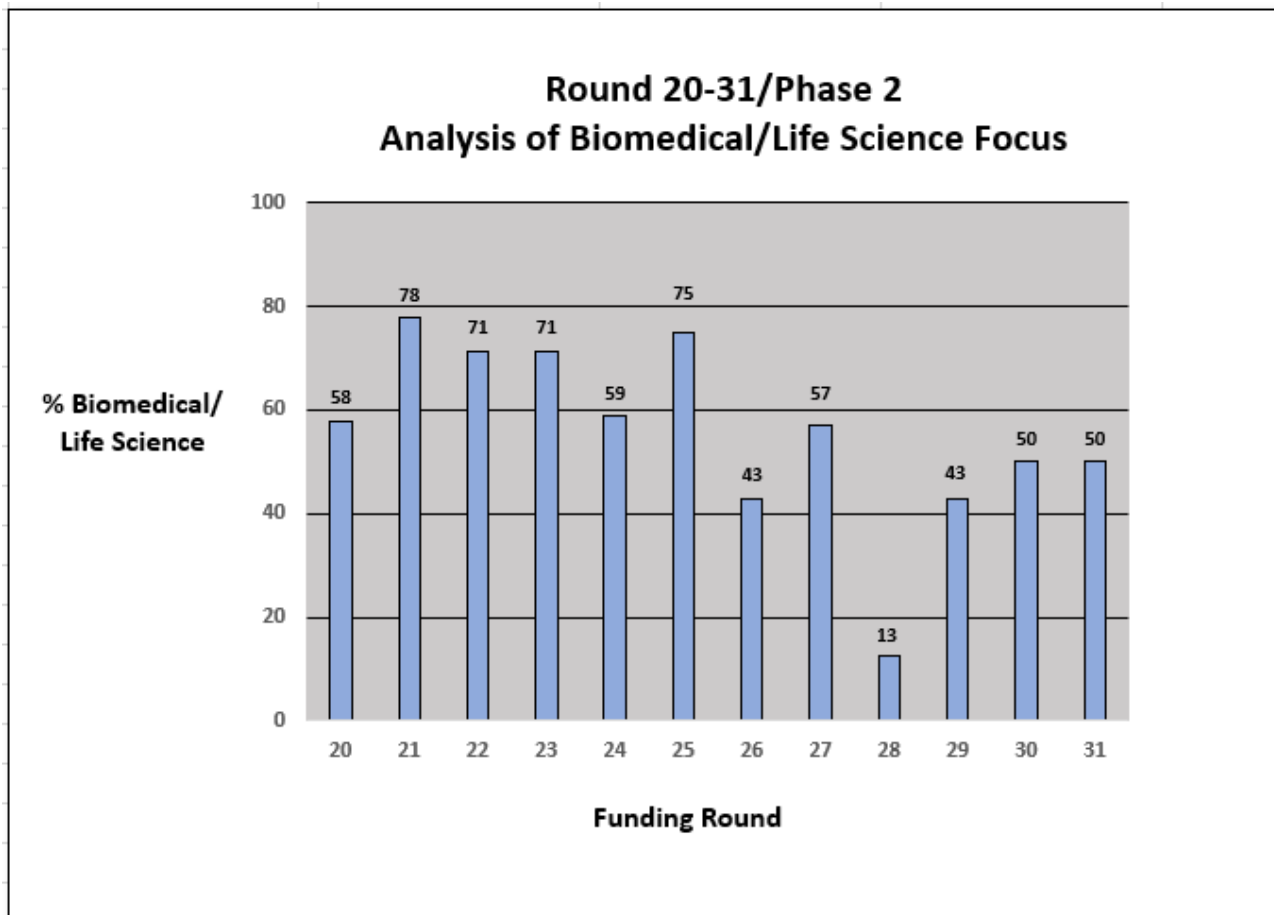


Figure 5 shows the percentage of Biomedical/Life Sciences applications for the last 12 Rounds. Round 31 represents 50% in Third Frontier Technology areas that are Biomedical/Life Sciences. Biomedical/ Life Sciences has been in the minority of the applications 3 times in the last 12 rounds. All twelve rounds average 56% of the applications in Biomedical/Life Sciences.

Figure 5: Rounds 20-31 Phase 2 Analysis of Biomedical/Life Science Focus



Carry Through and Reapplication

Phase 1 Carry Through: There were no Phase 2 applicant that previously received Phase 1.
There is one Phase 2 reapplication for the first time and it is recommended for funding.

5) Recommendations

Biomedical/ Life Sciences applications have been 50% or more of the applications in 9 of the last 12 rounds. Three of the last 6 rounds have had a minority of Biomedical/Life Sciences applications. It is recommended that active outreach efforts be developed to encourage more proposals in other Third Frontier Technology areas that reflect the diverse markets and economic activities in the State.

Appendix I

Summary of Redwood team and qualifications

Redwood, as a company, has been providing technology commercialization services for over 9 years while each team member has been active in this field for over 25 years.

Each Redwood team member

- possesses an advanced technical degree and extensive business proficiency
- has worked across the spectrum of technology commercialization from invention to successful market introduction
- understands how to assess a concept case from the perspective of aligning technologies to product applications in specific markets
- has lived, both conceptually and literally, the iterative process of understanding market needs and wants, value chains and who the customers are within the value chain

Team members have all worked for major corporations, research institutions, venture capital firms and technology start-up companies gaining a comprehensive understanding of what is necessary for development teams to successfully commercialize a technology. The Redwood team has served as evaluators for the Ohio Advanced Manufacturing program and an individual team member served as an evaluator for CALF, TIP and IOF loan programs for over a decade.

The five members of the Redwood team are highly qualified evaluators for the TVSF program and have combined experience and expertise in the following areas (combined years):

Commercializing technology into market pulled products (125+ years)

Market/Technology Assessment (140+ years)

Startup/ Spin out companies (50+ years)

Board member/Advisor to Startups (30+ years)

Evaluating/ monitoring RFPs/ Funding selection (40+ years)

The following is a brief summary of the five principal team members used in this evaluation Round.

Herb Bresler

- BS Biological Sciences, University of Maryland; BS Secondary Science Education, University of Maryland; PhD Immunology and Infectious Diseases, The Johns Hopkins University School of Hygiene and Public Health
- Former Senior Research Leader and Chief Scientist for Health and Life Sciences, Battelle Memorial Institute, responsible for evaluation of new technology-based business opportunities, intellectual property development, licensing and tech transfer; created and implemented new metrics to increase returns on discretionary R&D; cultivated approximately 1150 invention disclosures, 900 patent applications, and 120 granted patents, leading to \$52 million company funding
- Recipient of four R&D 100 awards for breakthrough medical devices in neuroscience and diagnostics
- Former Director of the Laboratory of Cellular Immunotherapeutics at the Arthur G. James Cancer Hospital and Research Institute at The Ohio State University

John McArdle

- BE, Manhattan College, MS, Northeastern University, Chemical Engineering
- MBA, Finance / International Business, University of Chicago (Booth School of Business)
- Former Business Development Manager, Battelle
- Former Product Line Manager – Koch Industries
- Former Technical Sales Manager, Allied Signal Corporation
- Recognized expert in water and wastewater treatment technologies
- Successful track record of introducing innovative technologies for a variety of municipal, industrial, and military applications in domestic and overseas markets.

Jim Sonnett

- BS, University of Virginia, MS, University of Massachusetts, PhD, University of Delaware, all in chemical engineering
- Former Vice President – Science and Technology, Battelle Health & Life Sciences
- Former R&D Leader – W. L. Gore & Associates and E. I. DuPont
- Built and led high impact innovation organizations in aerospace, electronics, and life sciences
- Former Board Member – Velocys, Ventaira, Battelle Ventures
- Adjunct professor – Ohio State University Fisher School of Business

Susan Stanton

- BS, Millersville University, Chemistry, MPH, Syracuse University, Organic Chemistry, PhD, University of Rochester, Organic Chemistry
- Personally developed 12+ products and led new product development teams at Mobay, Alcoa & Nexicor
- Holder of 10+ patents
- Former VP Market and Technology Assessment at the National Technology Transfer Center
- Over 15 years as an angel investor in technology-based startups
- Over 15 years as an evaluator for Ohio Third Frontier funds including IOF, CALF and TIP
- Over 8 years teaching market and business analytics to STEM graduate and post doc students.

Bhima Vijayendran

- BS, University of Madras, MS, University of Madras, PhD, University of Southern California in Polymer and Surface Science, MBA, University of New Haven
- Former Senior Research Leader and Vice President Business Development, Battelle Memorial Institute; Chief Research Officer, Battelle Science and Technology, Malaysia
- Former Director, Discovery Research, PPG Industries
- Recognized as one of the leading authorities on advanced materials, special chemical and polymer systems in numerous markets including: Renewable and clean technology, Energy, Nano Technology and Industrial Products.

Recipient of ten R&D 100 awards and over 100 patents and numerous other awards.

Appendix 2

TVSF objectives and phases

The Technology Validation and Start-up Fund (TVSF) provides grants under two phases to transition technology from Ohio Eligible Research Institutions into the marketplace through Ohio start-up companies. Under Phase 1, Ohio Research Institutions may apply for a pool of funds to support validation/ proof that will directly impact and enhance both the commercial viability of their unlicensed technologies and ability to support a start-up company. Under Phase 2, Ohio start-up and young companies may apply for funding to commercialize a technology they intend to license from a university or an Ohio research institution.

The goals of Phase 1 include:

- Generate the proof needed to move technologies to the point that they are either ready to be licensed by an Ohio start-up company or deemed unfeasible for commercialization. The institutions are encouraged to work with potential Ohio licensees to identify the proof needed.
- Perform validation activities such as demonstration and assessment of critical failure points in subsequent development, prototyping, scale-up and commercialization in order to generate this proof with strong preference for these activities being performed by an independent 3rd party source.

The goals of Phase 2 include:

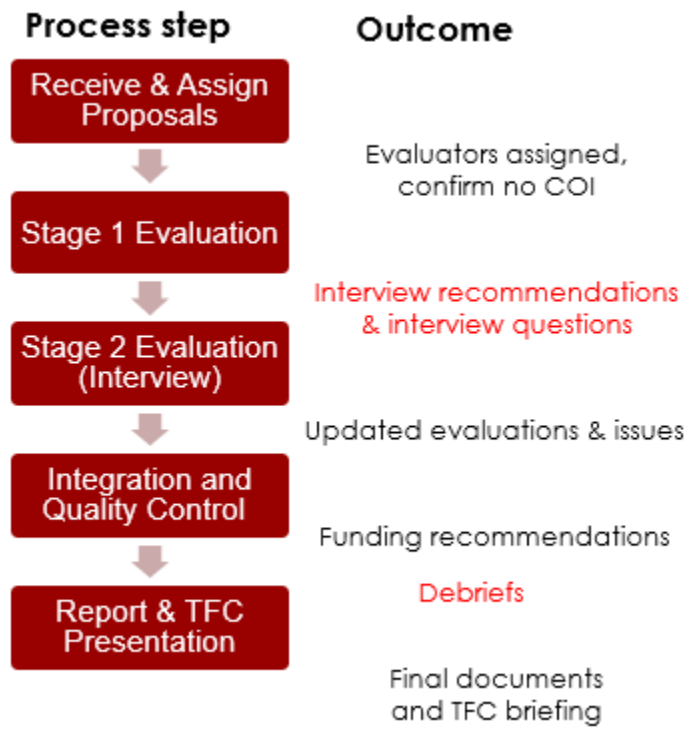
- Accelerate the commercialization of technology by Ohio start-up companies that license technology developed at Eligible Institutions during the critical early stage of life of the company.
- Generate the proof needed to move technology to the point where it is able to be commercialized or additional funds for commercialization can be raised. A clearly identified path to subsequent funding opportunities and working directly with potential investors to define the proof needed for investment into the company is strongly encouraged.
- Funded activities may include, but may not be limited to, beta prototype development and deployment to potential customers for testing and evaluation and market research/ business development in order to generate the proof needed.

Based upon these goals, the proposal evaluation criteria were developed. The proposals were then evaluated based on the criteria.

Description of review process

Review summary. Our overall review process flow and outcomes by stage are shown in Figure 1. A similar process has been successfully used by Redwood in prior projects for public and private clients. Discussions were held with the TVSF program manager after all but the initial step in Figure 1.

Figure 1. TVSF Evaluation Process



Review and Assign Proposal In this first step proposals were summarized and a primary evaluator was assigned who has the appropriate background and no conflict of interest.

Stage 1 Evaluation Stage 1 evaluations were conducted for each proposal using the criteria shown below in Tables 1 and 2. Differentially weighted criteria were used to evaluate Phase 1 and Phase 2 proposals. Each proposal was rated on a 0 (absent) – 5 (Outstanding) scale for each criterion, an approach used by the NSF and in other State of Ohio programs. The weightings reflect the experience of the Redwood team and our belief that some factors, for example team and market opportunity in Phase 2, are more important than others.

The entire review team subsequently discussed all the evaluations to ensure consistency and agreed upon which applicants to invite for interviews. Interview questions were then provided in advance to each applicant.

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Stage 2 Evaluations (Interviews) The standard procedure for this step is: In-person or Zoom (due to Covid restrictions), 45-minute interviews were held with each invited applicant to discuss the advance questions plus other topics of interest to the evaluators. A minimum of two Redwood team members participated in the interviews in person or Zoom with additional team members joining via conference call or Zoom. Interviews in this round were held via Zoom video conference call.

Integration and Quality Control Proposal evaluations were updated based on interview results. A calibration review was held by the review team to ensure that evaluations were performed consistently and that any changes made were a result of team consensus. Based on this review, proposals were recommended for funding.

Table 1 – Phase 1 Evaluation Criteria

Criterion	Weighting	Description
Alignment and Compliance	Go / No go	Institutional alignment with TVSF intent and compliance with RFP
Project Selection Committee	20	Skills, background and commitment of the committee members
Deal Flow; Budget Strategy	15	Is the projected deal flow consistent with the requested budget to enable committing funds within 1 year?
External Participation	15	Does process ensure validation activities will be performed by 3 rd parties; ESPs and state-funded programs/organizations are enlisted to enhance commercialization activities of the project?
Track Record	15	Is there a strong Phase 1 or comparable program track record of licensing and newco creation? If not, is there a plan for improvement?
Metrics	15	Realism and impact of proposed metrics, including licensing, start-ups.
Project Management & Experience	15	Is there a strong project management strategy and appropriate experience of people who allocate the pool of funds and manage individual projects?
Project Selection Process	5	Is there a clear, appropriate process for project selection?

TECHNOLOGY VALIDATION AND STARTUP FUND

Table 2 – Phase 2 Evaluation Criteria

Criterion	Weighting	Description
Alignment & compliance	Go / No Go	Proposal alignment with TVSF intent and compliance with RFP
Management Team	20	Skills, background and commitment
Opportunity / Market size	15	What is the market segment and total addressable market? Is it a platform or breakthrough technology or incremental improvement? If breakthrough, is it compatible with viable commercialization pathways?
IP Protection	15	Is IP adequately protected, does it enable the business model, is it differentiated from likely competition?
Compelling Proof of Concept	15	Was meaningful input from potential customers and key performance metrics used to design Proof of Concept? Are the competitive advantages compelling for potential customers?
Potential Investor / Business Partner Engagement	10	Is there company engagement / collaboration independent of licensing institution, including financial backing?
Business Model	10	Is the business model realistic AND achievable? Can the service / manufacturing model be scaled?
Project Plan / Budget Narrative	5	Is the budget consistent with proof in 1 year?
Growth Plan in Ohio	5	Does a start-up exist or is it planned? What are the plans to grow the company?
ESP Interaction	5	Is team engaged with ESP? Has team incorporated feedback from ESP into the project, proposal or business plan?